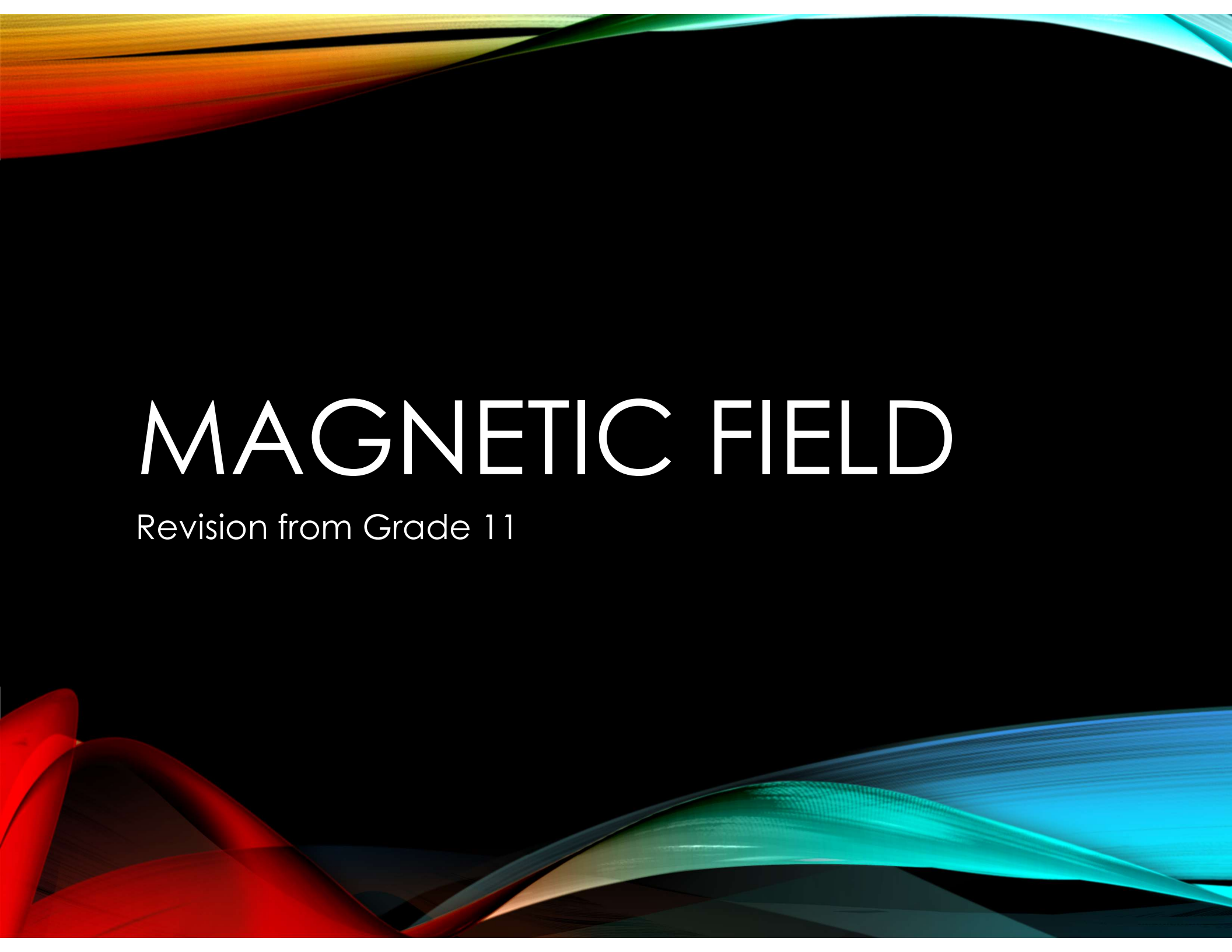


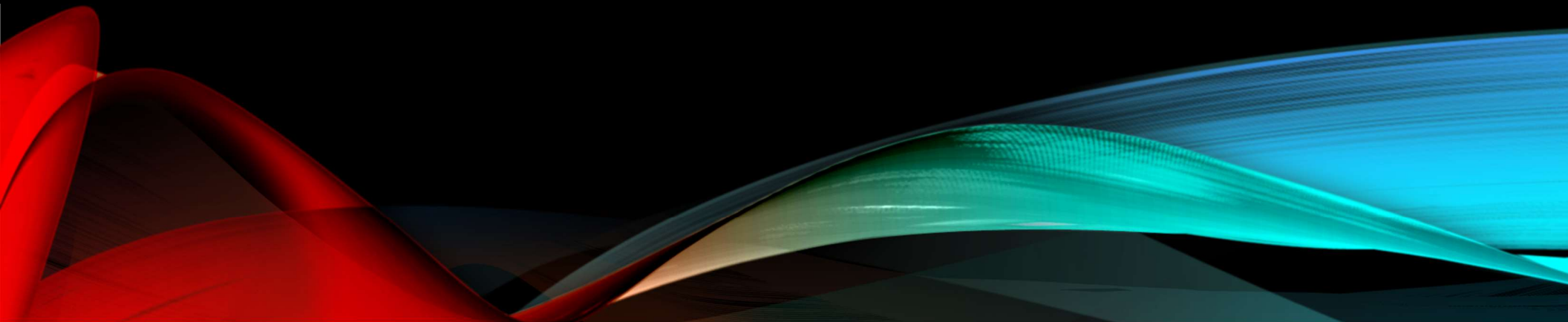


MAGNETIC FIELD

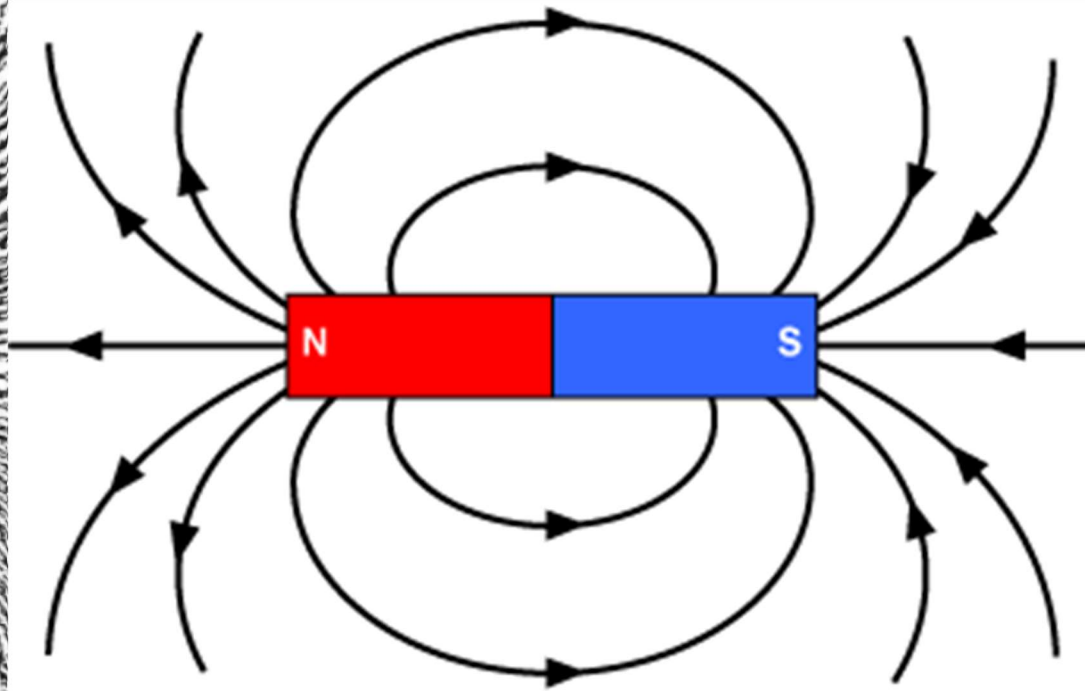
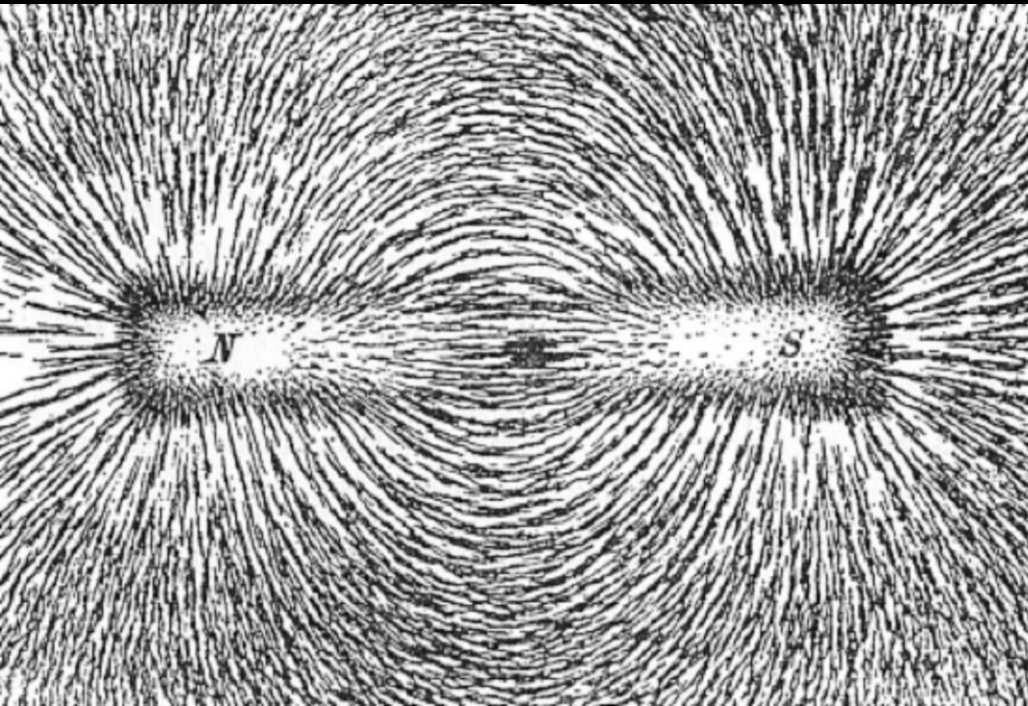
Revision from Grade 11

SECTION 1

Magnetic Fields around Permanent Magnets



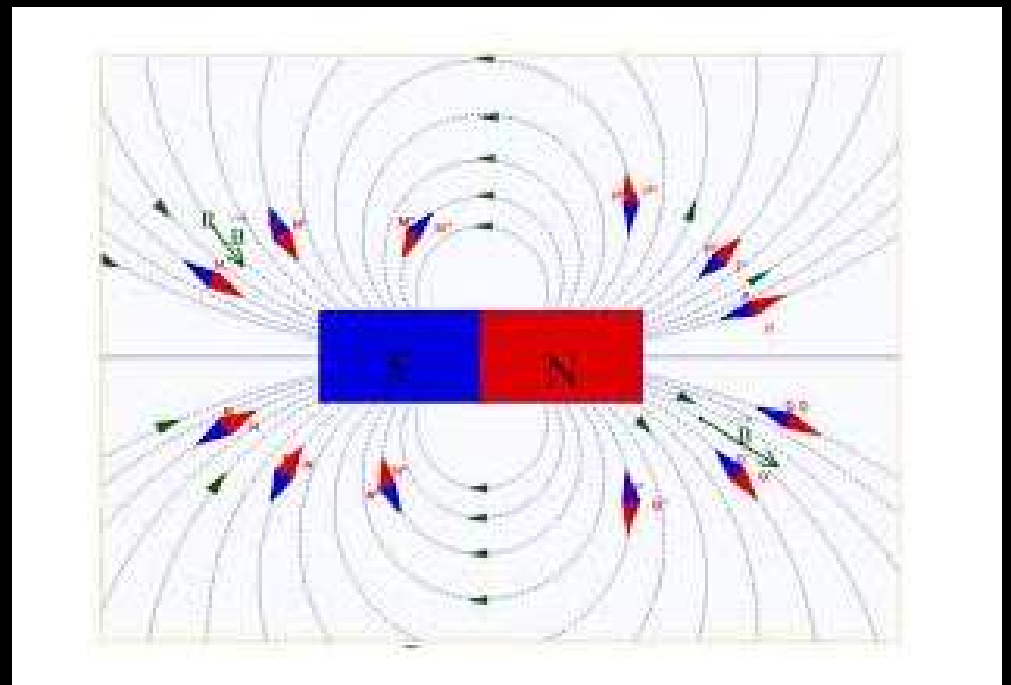
FIELD AROUND A BAR MAGNET



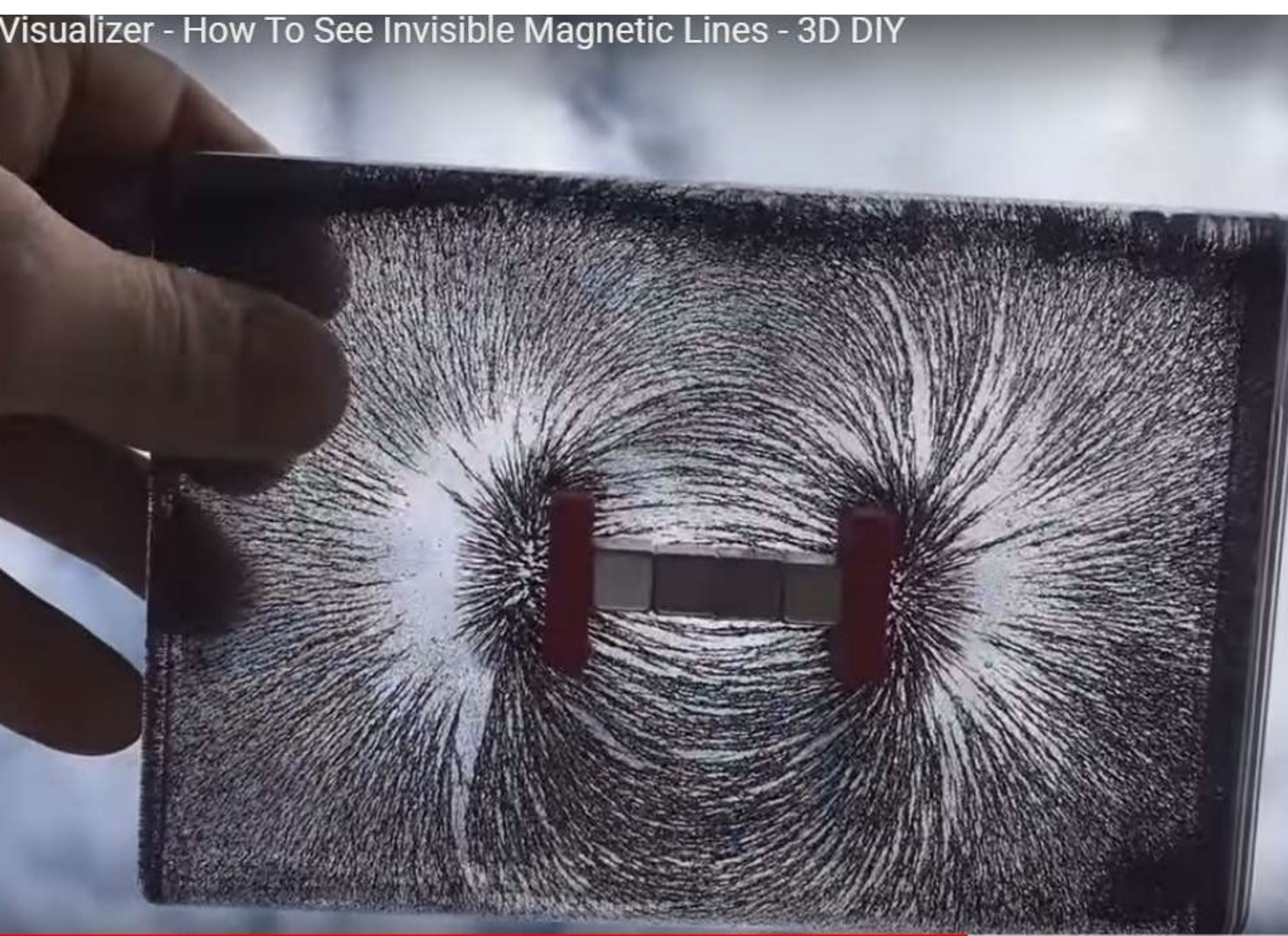
FIELD AROUND A BAR MAGNET

The magnetic field lines leave the north pole of a magnet, turn around and enter through its south pole.

These lines are the closest at the poles, where the magnet exerts its greatest effect.



Visualizer - How To See Invisible Magnetic Lines - 3D DIY

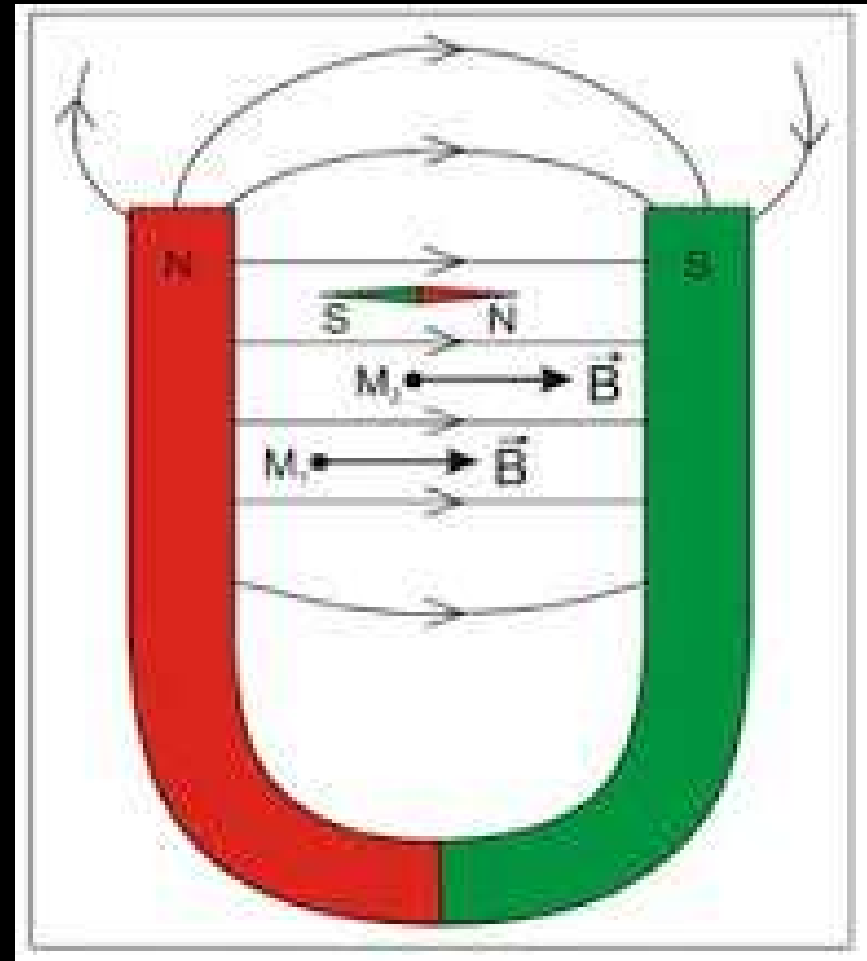


Field around a U-shaped magnet

Inside a U-shaped magnet, the magnetic field $\vec{B}$ is uniform:

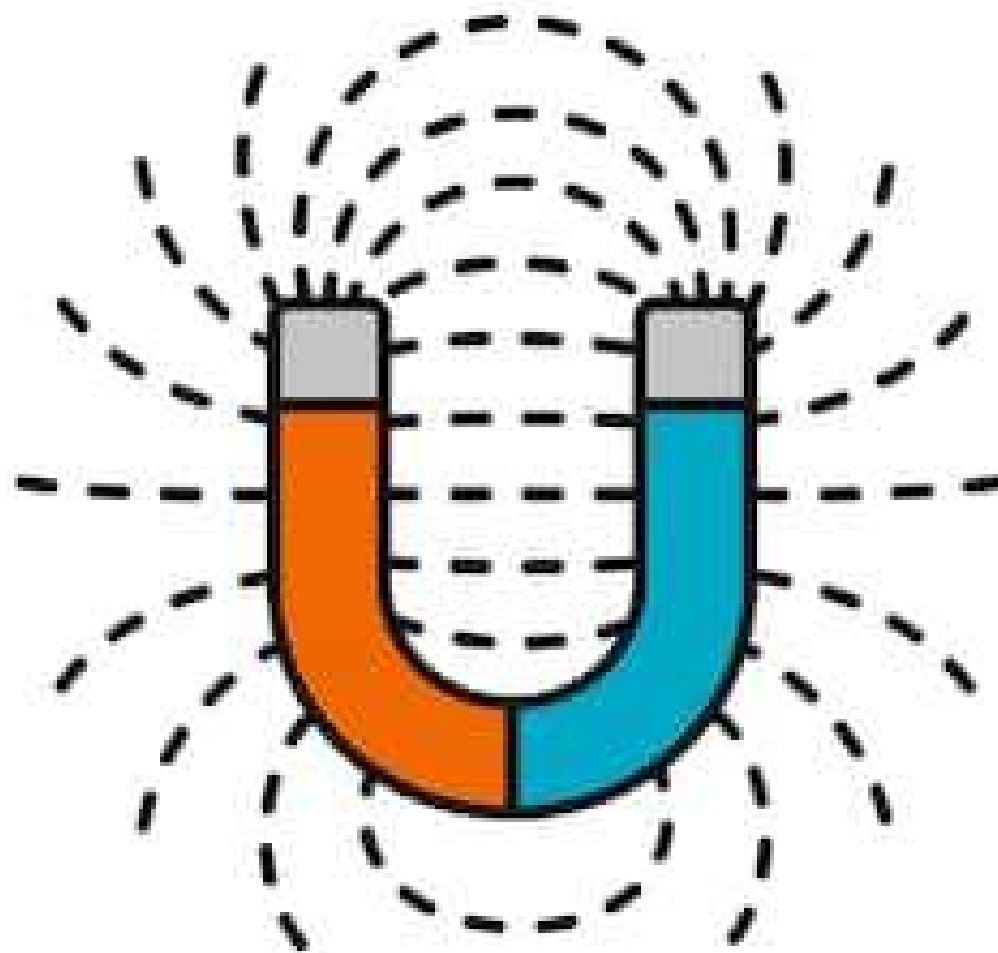
- it has constant direction and magnitude.
- Magnetic field lines are parallel to each other.

Again, the magnetic field $\vec{B}$ leaves the north pole of the magnet and enters through its south pole.



Field around a U-shaped magnet

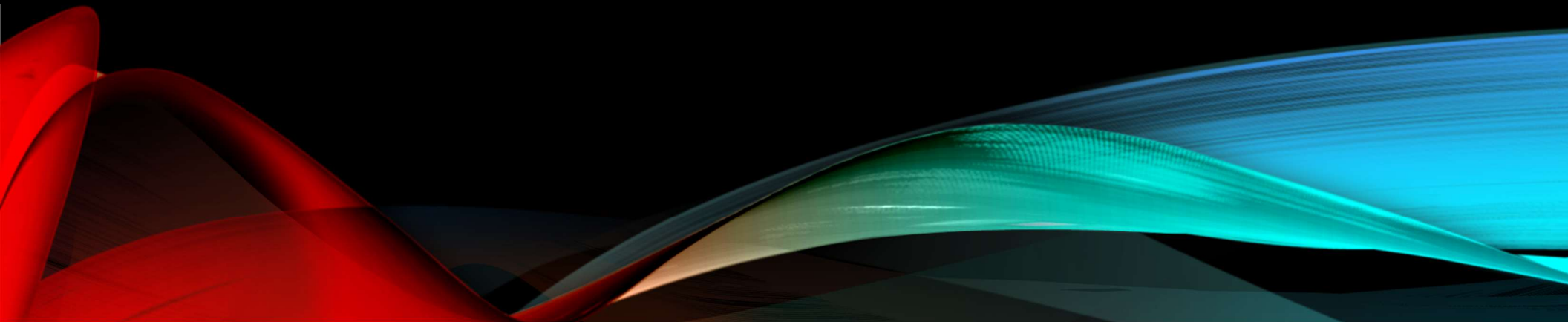




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SECTION 2

Magnetic Field around a current carrying wire



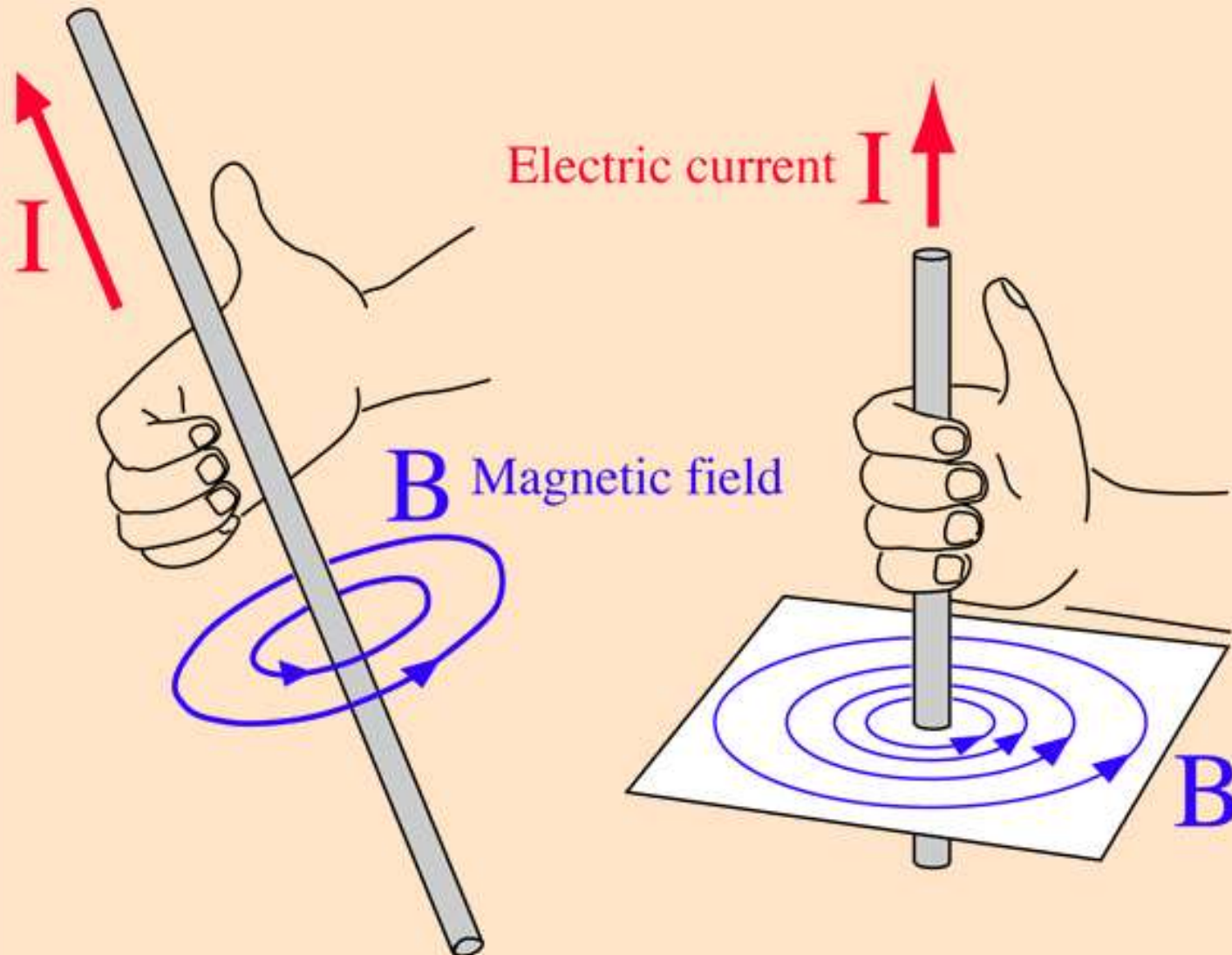
Field around a straight conductor

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Magnetic Field of Current

The magnetic field lines around a long wire which carries an electric current form concentric circles around the wire. The direction of the magnetic field is perpendicular to the wire and is in the direction the fingers of your right hand would curl if you wrapped them around the wire with your thumb in the direction of the current.



Field around a straight conductor



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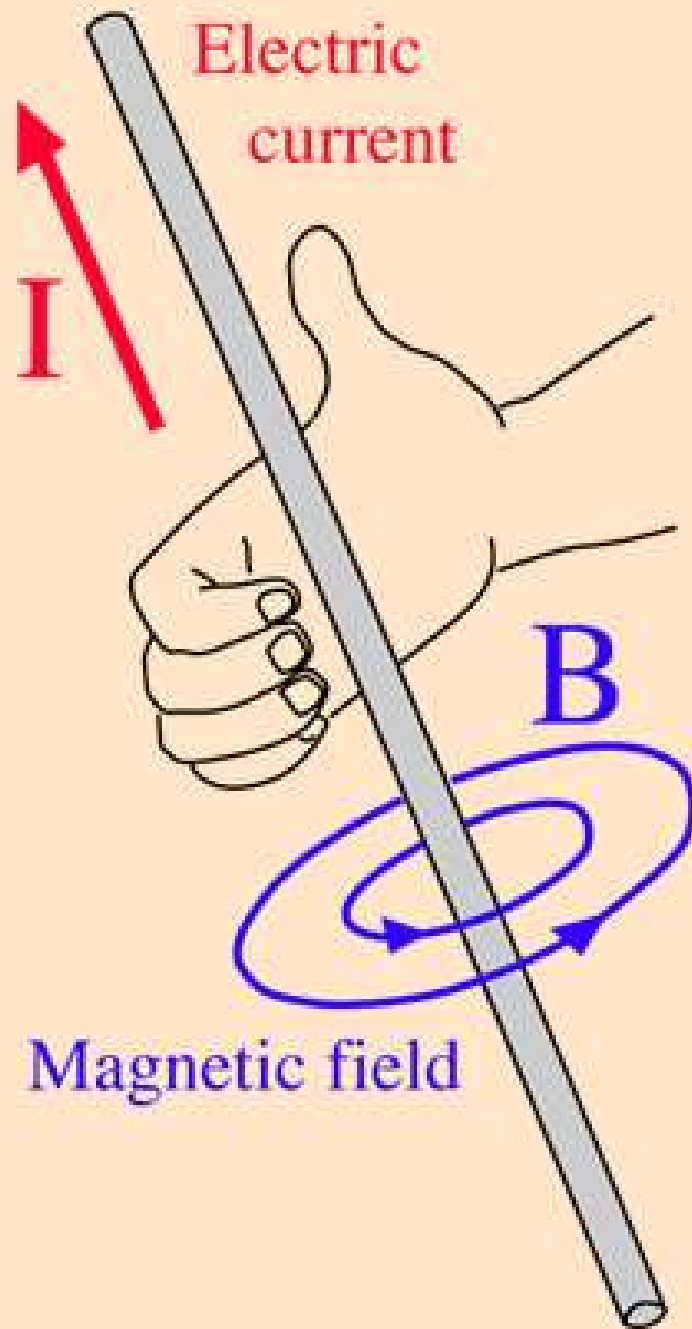


Field around a straight conductor

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Magnetic Field of Current



The magnetic field of an infinitely long straight wire can be obtained by applying Ampere's law. The expression for the magnetic field is

$$B = \frac{\mu_0 I}{2\pi r}$$

[Show](#)

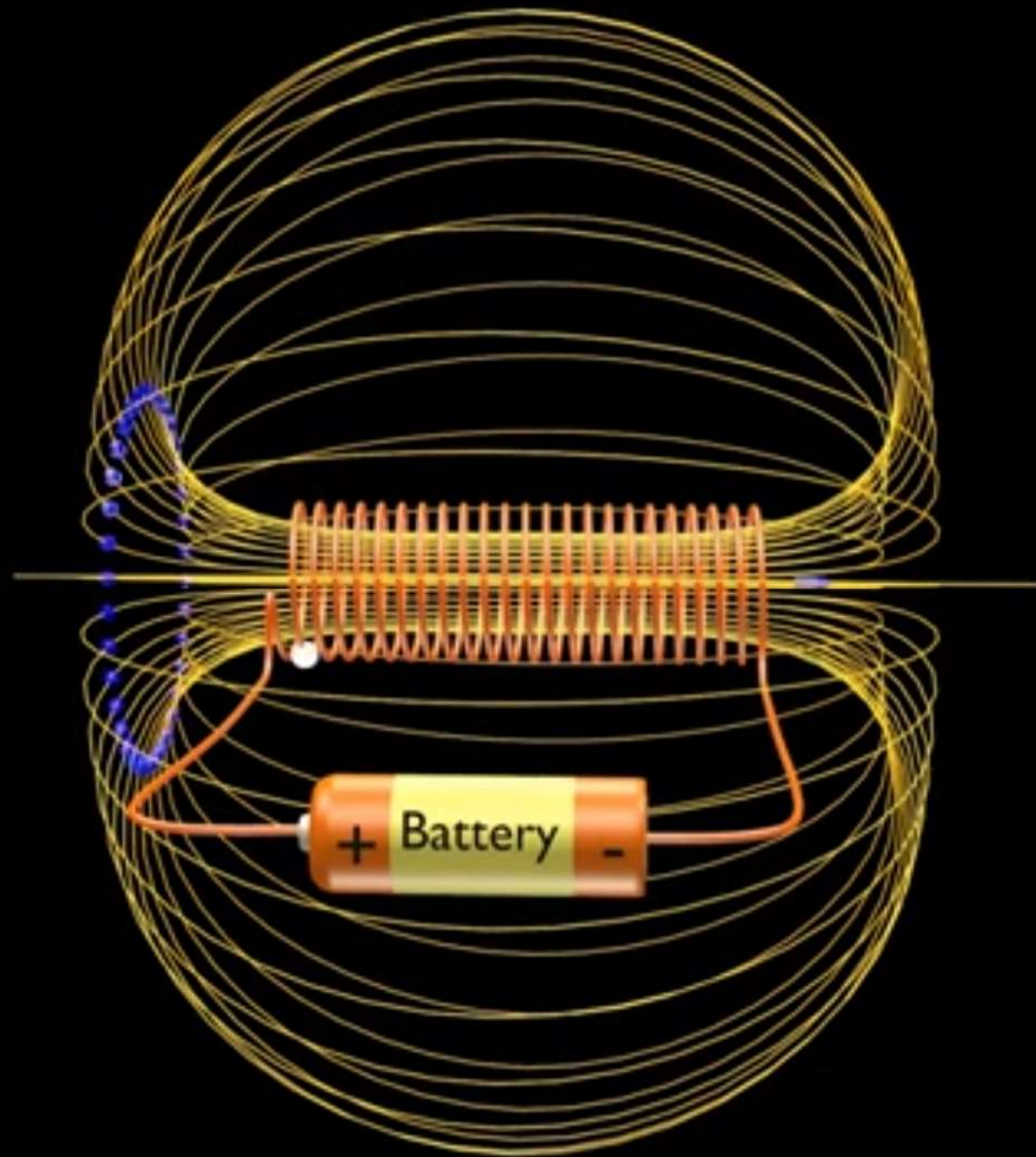
For a current $I =$ Amperes and radial distance $r =$ m, the magnetic field is

$$B =$$
 Tesla = Gauss.

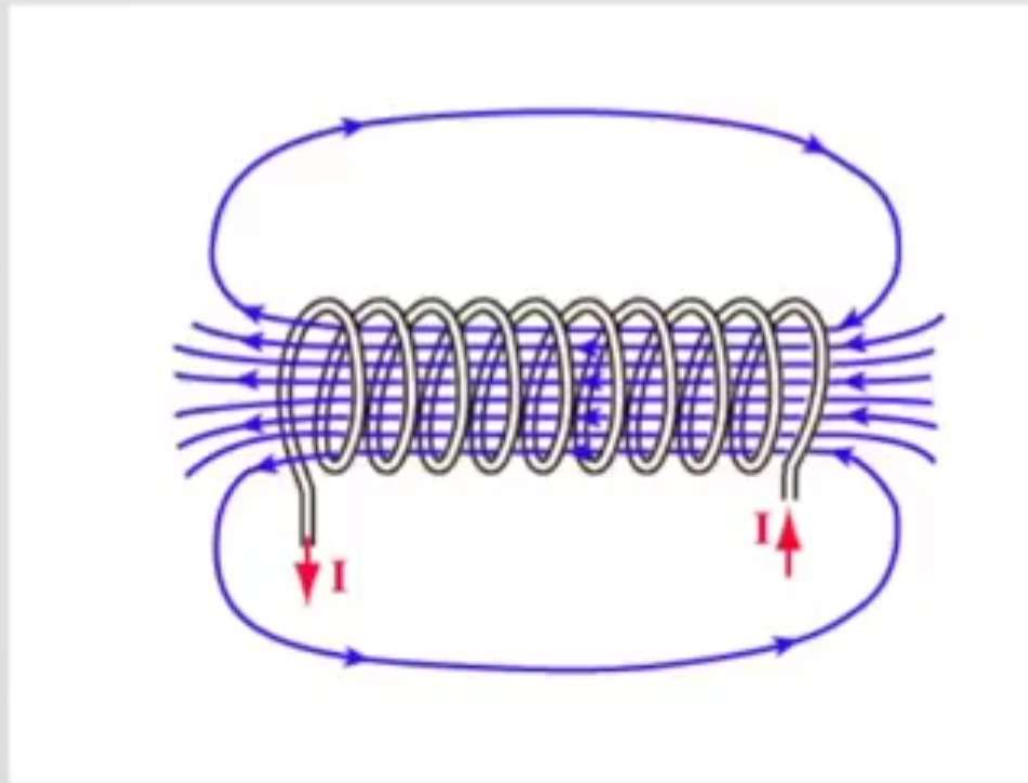
The earth's magnetic field is about 0.5 gauss. The permeability of free space is

$$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m} / \text{A}$$

Field around a coil/solenoid



Field around a coil/solenoid



Field around a coil/solenoid

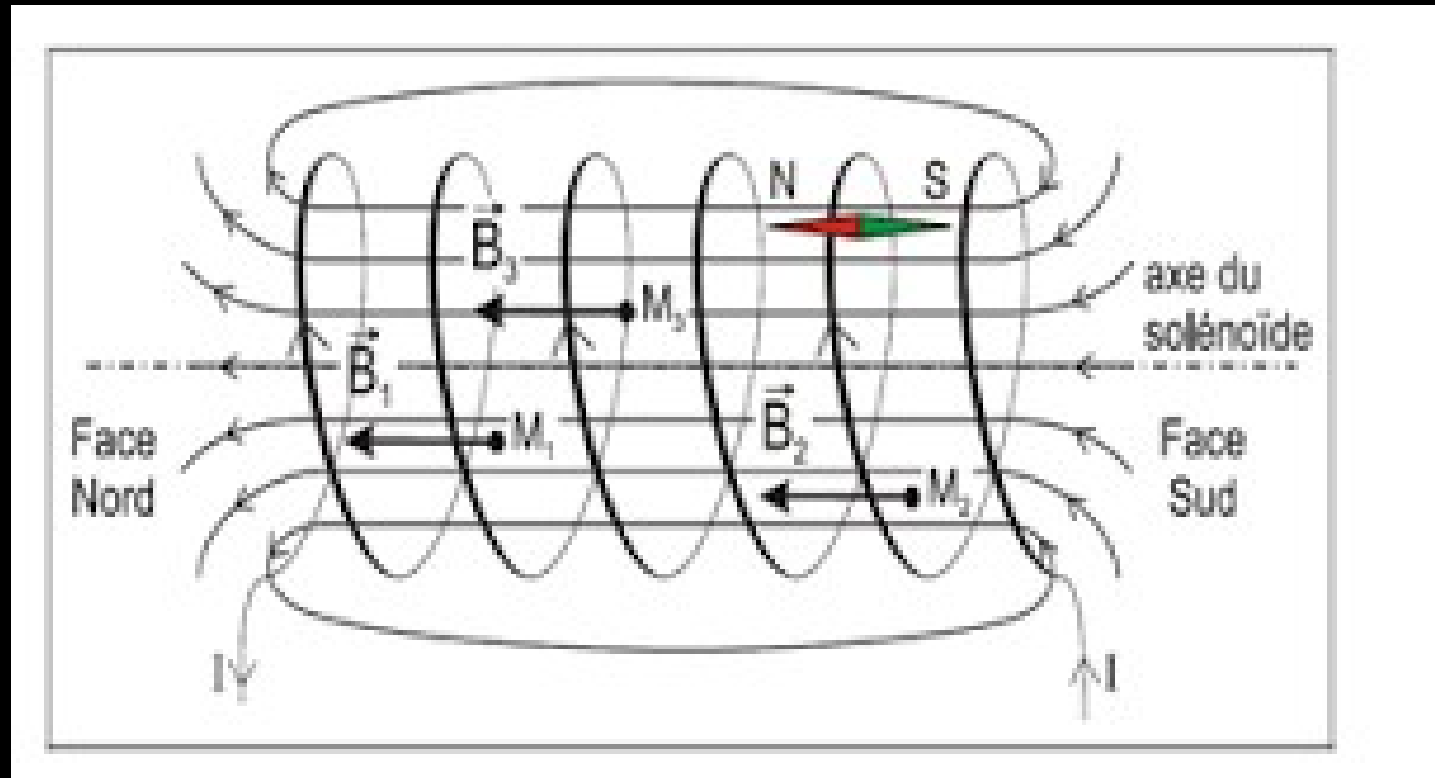


A solenoid (S) crossed by an electric current of intensity i produces a magnetic field $\vec{B}$. Inside the solenoid, the magnetic field is uniform. The magnetic field lines are parallel.

Characteristics of $\vec{B}_S$:

- Line of action: parallel to solenoid axis
- Direction: given by right hand rule
- Magnitude:

$$B_S = \mu_0 \frac{N \cdot i}{L}$$



$\mu_0 = 4\pi \cdot 10^{-7}$ (SI) : magnetic permeability of vacuum.

i : intensity of electric current in A.

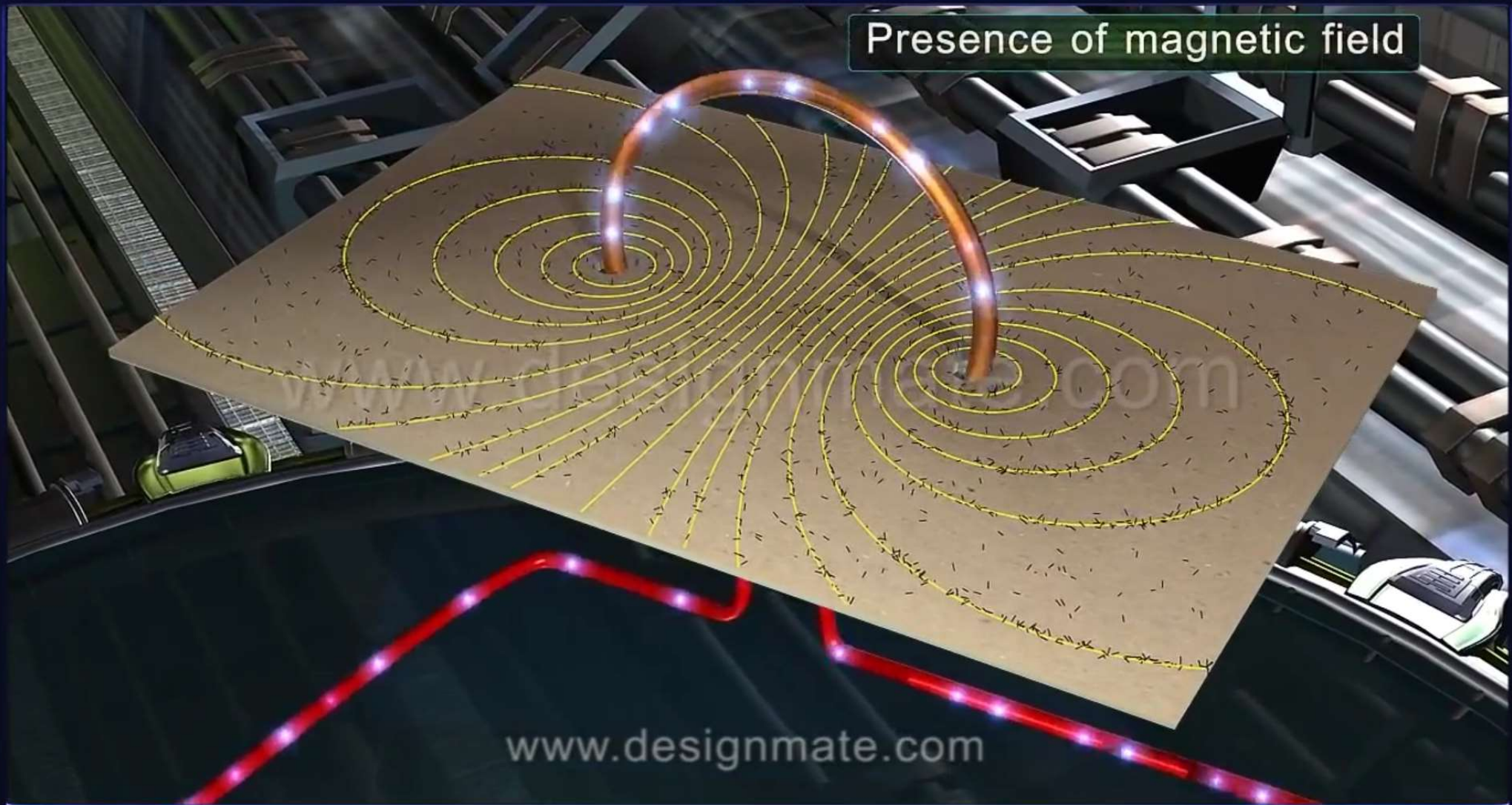
L : solenoid length in m.

N : is the number of turns of (S)

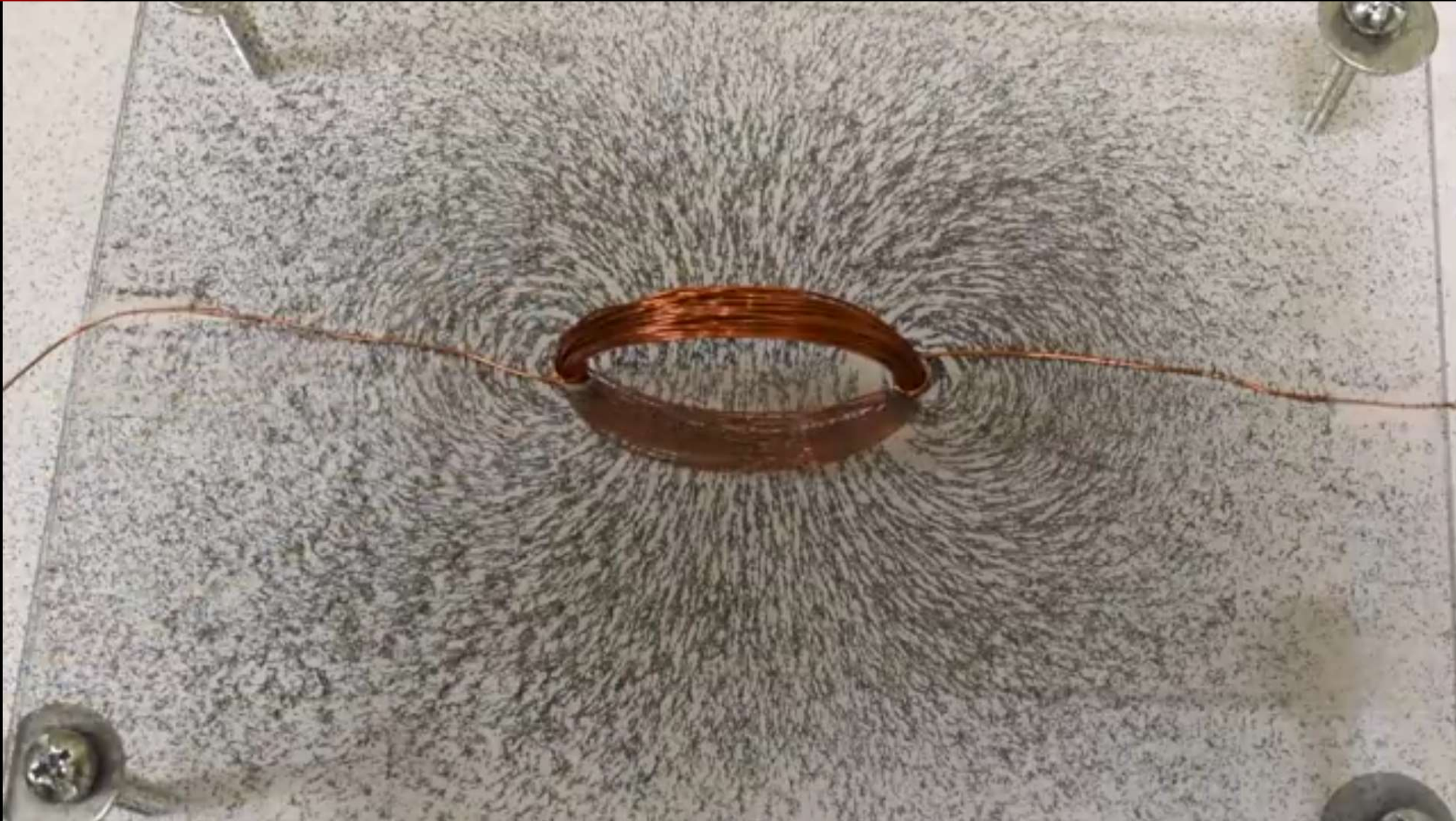
Field around a loop/flat coil

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Presence of magnetic field



Field around a loop/flat coil





For a flat coil made of N loops, the magnitude of the magnetic field at the center:

$$B_c = \frac{\mu_0 N \cdot i}{2R}$$